

C647

Trigonometry and Precalculus

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

Method © 2026 Erik Daniel Lindquist — Course Mastery Coach, armored drill. Items drafted with AI assistance under his direction.

How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A bicycle wheel from a shop that sells 400 bikes a year has a radius of 30 cm and turns through 120° . The rider is training for a charity ride. How far does a point on the rim travel?

- A. 60π cm
- B. 3,600 cm
- C. 40π cm
- D. 20π cm (about 62.8 cm)

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

An arc of length 15 cm lies on a circle of radius 5 cm. What is the central angle?

- A. 3 radians
- B. 75 radians
- C. 3 degrees
- D. $1/3$ radian

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A surveyor who has worked for the county for 11 years needs $\sin 210^\circ$ for a boundary calculation filed in triplicate. What is $\sin 210^\circ$?

- A. $\sqrt{3}/2$
- B. $-1/2$
- C. $-\sqrt{3}/2$
- D. $1/2$

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4. [TRANSFER — reversed stem / different variable]

$\cos \theta < 0$ and $\tan \theta > 0$. In which quadrant is θ ?

- A. Quadrant II
- B. Quadrant IV
- C. Quadrant I
- D. Quadrant III

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

A contractor building a wheelchair ramp for a library that reopened last fall uses a 20-foot ramp at a 10° angle to the ground. The library's budget was approved by the city. How high does the ramp rise?

- A. About 3.47 feet
- B. About 3.53 feet
- C. About 19.70 feet
- D. 200 feet

Target word _____	The one fact that decides _____	Answer _____	☐ sure ☐ unsure ☐ guess
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6.

An engineer at a firm that designs stadium lighting knows $\sin \theta = 3/5$ with θ in Quadrant II. The project is due in March. What is $\cos \theta$?

- A. $-3/4$
- B. $2/5$
- C. $-4/5$
- D. $4/5$

Target word _____	The one fact that decides _____	Answer _____	☐ sure ☐ unsure ☐ guess
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7.

A sound engineer at a studio that records 50 albums a year models a tone as $y = 3 \sin(4x) + 1$. The studio's walls are foam-lined. What is the period?

- A. 8π
- B. 4
- C. $\pi/2$
- D. 2π

Target word _____	The one fact that decides _____	Answer _____	☐ sure ☐ unsure ☐ guess
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8. [TRANSFER — reversed stem / different variable]

A sine function $y = \sin(Bx)$ has period 6π . What is B ($B > 0$)?

- A. $1/3$
- B. 6
- C. 3
- D. 6π

Target word _____	The one fact that decides _____	Answer _____	☐ sure ☐ unsure ☐ guess
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9.

An electrical engineer at a utility serving 90,000 homes writes $z = 2(\cos 30^\circ + i \sin 30^\circ)$. The utility recently upgraded its substations. What is z^3 ?

- A. $6i$
- B. $8(\cos 30^\circ + i \sin 30^\circ)$
- C. $8i$
- D. $8(\cos 33^\circ + i \sin 33^\circ)$

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10. [TRANSFER — reversed stem / different variable]

What is the product of $4(\cos 50^\circ + i \sin 50^\circ)$ and $3(\cos 20^\circ + i \sin 20^\circ)$?

- A. $12(\cos 30^\circ + i \sin 30^\circ)$
- B. $12(\cos 70^\circ + i \sin 70^\circ)$
- C. $12(\cos 1000^\circ + i \sin 1000^\circ)$
- D. $7(\cos 70^\circ + i \sin 70^\circ)$

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A theater that hosts a summer festival has 20 seats in the first row, and each row has 3 more seats than the row before. The theater was renovated in 2015. How many seats are in row 15?

- A. 20×3^{14}
- B. 45
- C. 65
- D. 62

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

A charity that runs a food bank receives \$12,000 in donations the first month, and each later month brings half as much as the month before. Its board meets quarterly. If this continued forever, what would the total donations approach?

- A. \$8,000
- B. \$24,000
- C. \$21,000
- D. It grows without limit

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer D Concept 1

A bicycle wheel ~~from a shop that sells 400 bikes a year~~ has a radius of 30 cm and turns through 120° . ~~The rider is training for a charity ride.~~ How far does a point on the rim travel?

Target word: arc length **Decides:** $120^\circ = 2\pi/3$ rad.

A	Confusion set	That is the full circumference, not a 120° arc.
B	Dropped element	Drops the conversion and multiplies 30×120 .
C	Wrong stem number	Uses the 60 cm diameter instead of the radius.
D	Correct	$30 \times 2\pi/3 = 20\pi$.

2. Answer A Concept 1 · transfer

An arc of length 15 cm lies on a circle of radius 5 cm. What is the central angle?

Target word: solve for the angle **Decides:** $\theta = s/r$ in radians.

A	Correct	$15 \div 5 = 3$.
B	Inverted operation	Multiplies $s \times r$.
C	Adjacent category	s/r gives radians, not degrees.
D	Confusion set	Divides r by s .

3. Answer B Concept 2

A surveyor ~~who has worked for the county for 11 years~~ needs $\sin 210^\circ$ for a boundary calculation ~~filed in triplicate~~. What is $\sin 210^\circ$?

Target word: $\sin$ of a QIII angle **Decides:** Reference 30° , sine negative in QIII.

A	Adjacent category	Uses cosine of the reference angle with the wrong sign.
B	Correct	$\sin 30^\circ = 1/2$, negative in Quadrant III.
C	Confusion set	That is $\cos 210^\circ$.
D	Dropped element	Drops the Quadrant III sign.

4. Answer D Concept 2 · transfer

$\cos \theta < 0$ and $\tan \theta > 0$. In which quadrant is θ ?

Target word: quadrant from signs **Decides:** Tan positive in I and III; cos negative in II and III.

A	Dropped element	Satisfies $\cos < 0$ but tan is negative there.
B	Inverted operation	Cos is positive in IV.
C	Confusion set	Cos is positive in I.
D	Correct	Only III has both.

5. Answer A Concept 3

A contractor building a wheelchair ramp for a library that reopened last fall uses a 20-foot ramp at a 10° angle to the ground. The library's budget was approved by the city. How high does the ramp rise?

Target word: height (opposite side) **Decides:** Opposite = hypotenuse $\times$ sin.

A	Correct	$20 \times \sin 10^\circ \approx 20 \times 0.1736 \approx 3.47$.
B	Adjacent category	Uses tangent as if 20 ft were the horizontal run.
C	Confusion set	Uses cosine, which gives the horizontal run.
D	Wrong stem number	Multiplies the length by the angle.

6. Answer C Concept 4

An engineer at a firm that designs stadium lighting knows $\sin \theta = 3/5$ with θ in Quadrant II. The project is due in March. What is $\cos \theta$?

Target word: cos from sin and quadrant **Decides:** $\cos^2 = 1 - 9/25 = 16/25$; negative in QII.

A	Confusion set	That is $\tan \theta$.
B	Inverted operation	Subtracts $1 - 3/5$ without squaring.
C	Correct	$\sqrt{(16/25)} = 4/5$, negative in Quadrant II.
D	Dropped element	Drops the Quadrant II sign.

7. Answer C Concept 5

A sound engineer at a studio that records 50 albums a year models a tone as $y = 3 \sin(4x) + 1$. The studio's walls are foam lined. What is the period?

Target word: period **Decides:** $2\pi / 4$.

A	Inverted operation	Multiplies 2π by 4.
B	Confusion set	Treats B as the period.
C	Correct	$2\pi \div 4 = \pi/2$.
D	Dropped element	Drops the effect of B.

8. Answer A Concept 5 • transfer

A sine function $y = \sin(Bx)$ has period 6π . What is B ($B > 0$)?

Target word: solve for B **Decides:** $2\pi/B = 6\pi$.

A	Correct	$B = 2\pi \div 6\pi = 1/3$.
B	Dropped element	Drops π from the period.
C	Inverted operation	Inverts the ratio.
D	Wrong stem number	Uses the period as B.

9. Answer C Concept 6

An electrical engineer at a utility serving 90,000 homes writes $z = 2(\cos 30^\circ + i \sin 30^\circ)$. The utility recently upgraded its substations. What is z^3 ?

Target word: power of a complex number **Decides:** DeMoivre: $2^3 \text{ cis } 90^\circ$.

A	Confusion set	Multiplies the modulus by 3 instead of cubing it.
B	Dropped element	Cubes the modulus but drops the angle change.
C	Correct	$8(\cos 90^\circ + i \sin 90^\circ) = 8i$.
D	Inverted operation	Adds 3 to the angle instead of multiplying.

10. Answer B Concept 6 • transfer

What is the product of $4(\cos 50^\circ + i \sin 50^\circ)$ and $3(\cos 20^\circ + i \sin 20^\circ)$?

Target word: product in polar form **Decides:** Multiply moduli, add angles.

A	Adjacent category	Subtracts angles, the rule for division.
B	Correct	$4 \times 3 = 12$; $50^\circ + 20^\circ = 70^\circ$.
C	Confusion set	Multiplies the angles.
D	Inverted operation	Adds the moduli.

11. Answer D Concept 7

A theater that hosts a summer festival has 20 seats in the first row, and each row has 3 more seats than the row before. The theater was renovated in 2015. How many seats are in row 15?

Target word: 15th term **Decides:** $a_{15} = 20 + 14 \times 3$.

A	Confusion set	Treats a constant difference as a ratio.
B	Wrong stem number	Multiplies 15×3 and drops the first row's 20.
C	Dropped element	Uses 15×3 , forgetting the first row has no added seats.
D	Correct	$20 + 42 = 62$.

12. Answer B Concept 8

A charity that runs a food bank receives \$12,000 in donations the first month, and each later month brings half as much as the month before. Its board meets quarterly. If this continued forever, what would the total donations approach?

Target word: infinite sum **Decides:** $a/(1 - r)$ with $r = 1/2$.

A	Inverted operation	Uses $a/(1 + r)$.
B	Correct	$12,000 \div (1 - 0.5) = 24,000$.
C	Dropped element	Adds only the first three months.
D	Absolute language	With $ r < 1$ the sum converges.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owled.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	9	
Wrong stem number	A figure copied from the prompt that is not the answer.	4	
Adjacent category	The right idea in the wrong bucket.	4	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	8	
Absolute language	Always, never, guarantee — where judgment applies.	1	
Confusion set	Every option is real; the test is exact role-matching.	10	
Camouflage	Answers a stem fact that was never the question.	0	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Radians and arc length	Arc length $s = r\theta$ only when θ is in radians; convert degrees with $\theta = \text{degrees} \times \pi/180$.	Plugging degrees into $s = r\theta$.	An angle in degrees and a radius with "how far along the arc".
2	Reference angles and signs by quadrant	Evaluate with the reference angle, then attach the sign for the quadrant: sine positive in I and II, cosine in I and IV, tangent in I and III.	Using the reference-angle value but forgetting the sign.	An angle past 90° .
3	Right-triangle ratios	$\sin = \text{opposite/hypotenuse}$, $\cos = \text{adjacent/hypotenuse}$, $\tan = \text{opposite/adjacent}$; choose the ratio that links the known side to the unknown.	Picking cosine or tangent when the hypotenuse and opposite side are involved.	A ramp, ladder, or line of sight with an angle.
4	Pythagorean identity	$\sin^2\theta + \cos^2\theta = 1$ gives one ratio from the other; the quadrant fixes the sign.	Subtracting without squaring, or dropping the sign.	"Given $\sin \theta$ and the quadrant, find $\cos \theta$."
5	Sinusoid transformations	For $y = A \sin(Bx - C) + D$, the amplitude is $ A $, the period is $2\pi/ B $, and D shifts the midline.	Treating B as the period or multiplying 2π by B .	"Period" or "amplitude" of a trig function.
6	Polar complex numbers and DeMoivre	In polar form, multiply moduli and add angles; $(r \text{ cis } \theta)^n = r^n \text{ cis}(n\theta)$.	Multiplying the modulus by n or adding n to the angle.	A complex number in trig form raised to a power.
7	Arithmetic sequences	$a_n = a_1 + (n - 1)d$.	Using $n \cdot d$ instead of $(n - 1)d$, or treating a constant difference as a ratio.	A starting value and a constant added each step.
8	Infinite geometric series	If $ r < 1$, the sum is $a_1/(1 - r)$; if $ r \geq 1$, it diverges.	Summing only the listed terms or saying all infinite sums diverge.	"Continues forever" with each term a fixed fraction of the last.

Course scope (public description): Trigonometry and Precalculus covers the knowledge and skills necessary to apply trigonometry, complex numbers, systems of equations, vectors and matrices, and sequences and series, and to use appropriate technology to model and solve real-life problems. Topics include degrees; radians and arcs; reference angles and right triangle trigonometry; applying, graphing and transforming trigonometric functions and their inverses; solving trigonometric equations; using and proving trigonometric identities; geometric, rectangular, and polar approaches to complex numbers; DeMoivre's Theorem; systems of linear equations and matrix-vector equations; systems of nonlinear equations; systems of inequalities; and arithmetic and geometric sequences and series. College Algebra is a prerequisite for this course.